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A Multiple Attribute Utility Theory Approach to Ranking and Selection

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Managers of large industrial projects often measure performance by multiple attributes. For example, our paper is motivated by the simulation of a large industrial project called a land seismic survey, in which project performance is based on duration, cost, and resource utilization. To address these types of problems, we develop a ranking and selection procedure for making comparisons of systems (e.g., project configurations) that have multiple performance measures. The procedure combines multiple attribute utility theory with statistical ranking and selection to select the best configuration from a set of possible configurations using the indifference-zone approach. We apply our procedure to results generated by the simulator for a land seismic survey that has six performance measures, and describe a particular type of sensitivity analysis that can be used as a robustness check.
(*Simulation; Ranking and Selection; Multiple Attribute Utility Theory*)

1. Introduction

In recent work, Morrice et al. (1997) developed a simulator of a project that contains multiple input parameters and multiple performance measures. This paper describes the application and details of how we use the simulator, multiple attribute utility theory, and statistical ranking and selection (R&S) to select the best project configuration of K (>1) possible configurations. The K configurations are constructed from different settings of the simulator input parameters.

Evaluating project configurations on multiple performance measures complicates the R&S analysis. Most of the R&S literature focuses on procedures that are designed for scalar performance measures (see, for example, Bechhofer et al. 1995). However, some multivariate results do exist. Gupta and Panchapakesan (1979, Chapter 7) describe procedures that are based on scalar functions of the mean and covariance matrix of the multivariate populations. In other words, these procedures reduce the multivariate performance measure problem to a scalar performance measure problem. Andijani (1998) uses the analytic hierarchy process (AHP) (e.g., Saaty 1988) to select the most preferred Kanban allocation. The paper provides an excellent example of the need for a method to tradeoff conflicting objectives when analyzing the output of the simulation of a system, including sensitivity analysis. Finally, Kim and Lin (1999) use a maximin approach and desirability functions to optimize the characteristics of steel.

In a business setting, a different approach is often used: Convert project performance over multiple measures to a scalar measure using costs. Although this type of "costing" or pricing out performance measures has many obvious advantages, it has some disadvantages as well. For example, accurate cost data may not be available because of insufficient resources. Additionally, it may be difficult to accurately attach

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a dollar value to intangibles (e.g., the quality of life, good will, etc.) even if the resources are available.

We propose another approach to this problem: Convert multiple performance measures to a scalar performance measure using multiple attribute utility (MAU) theory (for a thorough review of MAU, see Keeney and Raiffa 1976 and von Winterfeldt and Edwards 1986). MAU theory can be used instead of a costing approach when good cost data are not available or when cost is not suitable as a measure of performance. Alternatively, MAU theory can be used to embellish costing information that is considered to be incomplete (e.g., to account for the intangibles). In this paper, we combine MAU theory with statistical R&S using the indifference-zone approach. The goal is to select the best project configuration from a set of K configurations when project performance is determined by $n > 1$ performance measures. This work was introduced in Morrice et al. (1998) and further embellished in Morrice et al. (1999).

Our approach has a number of advantages. First, the use of MAU results in a *rigorous* "costing" of the other criteria; this procedure is also referred to as "making even swaps" (Hammond et al. 1998). Second, our approach closely relates to the common business practice of costing or assigning preferences to different performance measures. Third, unlike the procedures described in Gupta and Panchapakesan (1979), our method does not require the complicated step of estimating a covariance matrix. As a result, the R&S approaches that we use are simpler to implement and belong to a class of procedures whose properties are better understood. Fourth, a potential shortcoming of Andijani (1998) and Kim and Lin (1999) is that no attempt is made to determine if the number of simulations is sufficient to identify the top-performing configuration. By developing a formal R&S procedure, we provide an approach that estimates the number of simulations required to select the best configuration with a high level of probability. Finally, as Kim and Lin (1999) indicate, in some situations a maximin approach may lead to an unreasonable decision because it focuses on maximizing the poorest performing criterion. For this reason, they recommend performing "several approaches for the final decision" (Kim and Lin 1999, p. 8). Our approach will overcome these "unreasonable" decisions because it is a fully compensatory technique (see §5.4). In other words, our approach allows good performance on one criteria to compensate for poor performance on another, and we assess the *relative* importance of each criterion. Previous work on multivariate R&S has been limited by the lack of a trade-off mechanism that allows the decision maker to combine explicitly seemingly disparate performance measures. As described by Gupta and Panchapakesan (1979, p. 141),

> [W]e consider two bivariate normal populations with mean vectors $\mu_1 = (\mu_{11}, \mu_{12})$ and $\mu_2 = (\mu_{21}, \mu_{22})$ and a common covariance matrix Σ.... In this case we will naturally prefer the first population if $\mu_{1j} \geq \mu_{2j}$, $j = 1, 2$. However if $\mu_{11} > \mu_{21}$ and $\mu_{12} < \mu_{22}$, the two are not comparable. There is practically no result available in this direction.

One goal of this work is to rectify these shortcomings in the current R&S literature by providing a fully compensatory technique.

Of course, there are several other issues that must be recognized as critical for the success of our approach. Because we do use a fully compensatory model this requires elicitation of both a utility function for each measure as well as a "weight," or indication of relative importance. While techniques for handling these types of elicitations are readily available (e.g., Clemen 1991 and Keeney and Raiffa 1976), it is important to realize that care must be taken to ensure that the decision maker's preferences are accurately represented. We will present example results, as well as a mechanism for conducting a sensitivity analysis, to address this concern. In addition, the determination of an R&S indifference-zone parameter, δ^*, is complicated by use of MAU. Performance measures are converted from their natural units into "utils" that have little meaning to the decision maker. Choice of an invertible utility function, of which there are many that are commonly used in decision analysis, allows for a simple conversion from utils to meaningful units.

The idea for the methodology outlined in this paper is based on interactions with real decision makers whose decision inputs are the results of simulation experiments. When the standard ranking and selection procedures were recommended, the decision

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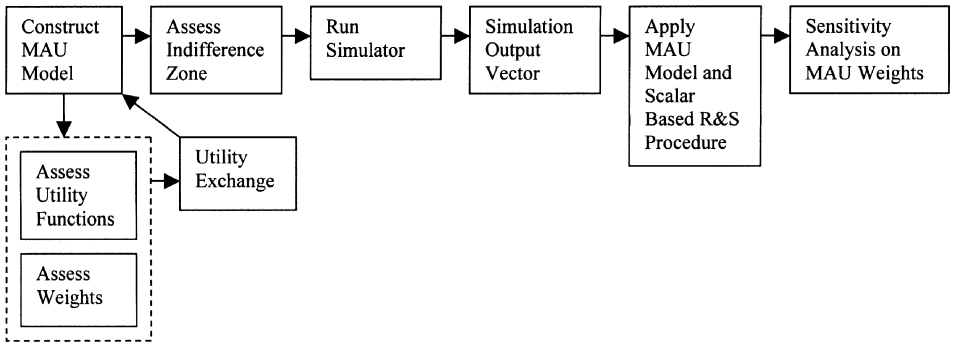
Figure 1 General Outline of the Procedure



makers responded with a question: "What if more than one thing is important?" While we certainly sought to develop a practical tool for this particular class of problem, we wanted a theoretically sound mechanism to provide this decision support. The result was a combination of two sound theories: multiattribute utility theory and ranking and selection. This synthesis extends the ranking and selection literature by accommodating multiple criteria, and provides another application area for multiattribute utility theory.

Figure 1 contains a general outline of our approach. A MAU model is constructed to analyze the vector of simulation results. Once an indifference zone has been established for the MAU model, a scalar R&S procedure is applied to the results. It is important to emphasize that the MAU model should be assessed, i.e., the analyst should assess the utility functions, weights, and the indifference zone parameter, *prior* to the running of the simulation models to avoid biasing the R&S. The final step is to conduct sensitivity analysis on the MAU weights to assess the robustness of the best configuration to uncertainty in the weights. It is important to note that a simulation model need not generate the data. In fact, the two boxes in Figure 1 representing simulation could be replaced by anything that generates statistical samples.

The remainder of the paper is organized in the following manner. Section 2 describes the project example from Morrice et al. (1997) that will be used throughout the paper. Section 3 contains a brief overview of the MAU theory, and §4 provides the setup for R&S and a description of the combined R&S and MAU procedures. Section 5 discusses the selection of the indifference zone parameter δ^*. Section 6 illustrates application of the procedures on the project example described in §2, and analyzes the robustness of the final recommendations via sensitivity analysis. Section 7 contains concluding remarks.

2. Case Example

We use the methodology developed in this paper to analyze the results generated by a simulator of the project described in Morrice et al. (1997). The simulator models a large outdoor operation called a land seismic survey. Land seismic surveys generate geophysical information used in oil and gas exploration (Dobrin and Savit 1988). They are conducted over large geographical areas (tens to hundreds of square kilometers). These projects take anywhere from a few days to a few years to complete, employ from 20 to 1,000 people, require capital equipment valued in the tens of millions of dollars, and generate survey revenues ranging from hundreds of thousands to hundreds of millions of dollars. The simulator was designed to support bidding, planning, and conducting these large, complicated, and expensive projects in a profitable manner.

The primary objective associated with modeling a land seismic survey is to characterize the various loosely coupled, interacting operations in a realistic manner. Achieving this objective is complicated by several factors: terrain, equipment failures, wildlife (causes damage to equipment), weather, permitting (i.e., permission to conduct the survey on public

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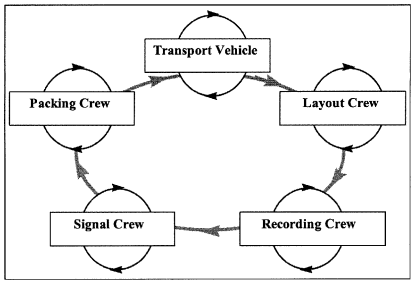
Figure 2 **Crews in a Land Seismic Survey**



or private property), personnel problems, community relations, cultural factors, and environmental and legal restrictions. Given the complexity of operations and the uncertainty associated with several of these factors, simulation was identified as an important decision support tool for use in understanding, quantifying, and improving land seismic survey operations.

The execution of a land seismic survey requires the coordination of five types of crews (see Figure 2). Briefly, the source crew sends signals (shock waves through the earth) from several geographic locations. The recording crew records reflections of these signals from the earth's subsurface layers. The layout crew places receiving (or monitoring) equipment at several geographic locations so that the recording crew can receive the reflected signals. The transport crew brings the layout crew receiving equipment. The packing crew prepares receiving equipment for the transport crew that is no longer required on a particular part of a survey for receiving signals sent by the source crew. Figure 2 summarizes the sequence above in an operations cycle diagram. The diagram shows that each crew cycles through its own local operations steps (e.g., the transport crew picks up and drops off equipment) depicted by small loops on each crew. Additionally, each crew forms a step in the larger operations cycle that represents the progression of the land seismic survey. For more details on land seismic survey of operations, see Morrice et al. (1997).

During an actual land seismic survey, a project manager will monitor multiple performance measures. These include project cost, project duration, and utilization for all types of crews. Project cost represents the bottom line and is considered the most important performance measure. Project duration, which is positively correlated with project cost since variable costs such as labor increase with the duration of the job, is included because certain things such as reputation for finishing the job in a timely manner may be difficult to price.

The crew utilization is monitored to ensure that crews are not overutilized. Crews work under very adverse conditions on many land seismic surveys. Overutilization of crews can lead to worker dissatisfaction, poor quality work, attrition, and unsafe working conditions. Again, things such as worker dissatisfaction and work quality may be difficult to cost. Therefore, these measures are monitored in addition to cost.

To mimic reality, our simulator generates statistics on cost, duration, and all crew utilizations. We use the simulator to compare different project configurations (e.g., different survey designs or different levels of resources). Since the evaluation of different project configurations must be based on multiple, possibly correlated, measures, we needed to develop a methodology to solve this problem. For more details on the simulator, see Morrice et al. (1997).

3. MAU Theory: An Overview

MAU theory (Keeney and Raiffa 1976) is one of the major analytical tools associated with the field of decision analysis (see, for example, Clemen 1991). A MAU analysis of alternatives (in our example, project configurations) explicitly identifies the measures that are used to evaluate the alternatives, and helps to identify those alternatives that perform well on a majority of these measures, with a special emphasis on the measures that are considered to be relatively more important.

The first step in a MAU analysis is to form a matrix in which each row corresponds to an alternative and each column represents a performance measure. The cells of the matrix contain estimates of the performance of each alternative on each of the measures. When these estimates are uncertain, it is often appropriate to quantify them with ranges or with probability distributions determined using risk analysis methods (e.g., Clemen 1991, Keeney and von Winterfeldt 1991), i.e., Monte Carlo simulation.

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Next, for each performance measure a single-attribute utility function is assessed that scales performance between 0 and 1, inclusive (von Neumann and Morgenstern 1947). A multiple attribute utility function determines how the performance on each measure affects overall performance vis-à-vis a set of assessed weights or measures of relative importance. There are many forms of MAU functions that are theoretically valid; however, the multilinear utility function (1a) is the most general form that is used with any regularity in applications of MAU,

$$u(\mathbf{X}) = \sum_{i=1}^{n} w_i u_i(X_i) + \sum_{i=1}^{n}\sum_{j>i} w_{ij} u_i(X_i) u_j(X_j) + \sum_{i=1}^{n}\sum_{j>i}\sum_{m>j>i} w_{ijm} u_i(X_i) u_j(X_j) u_m(X_m) + \cdots + w_{123\ldots n} u_1(X_1) u_2(X_2) \ldots u_n(X_n), \quad (1a)$$

where $\mathbf{X} = (X_1, X_2, \ldots X_n)$ is a vector of random variables over performance measures, $u_i(\cdot)$ is a single-attribute utility function over measure i that is scaled from 0 to 1, w_i is the weight for measure i where $0 \le w_i \le 1$ for all i, and w_{ijm} are scaling constants that represent the impact of the interaction between attributes i, j, and m on preferences, for example. In the remainder of this paper we will use the standard convention in the probability and statistics literature of representing a realization of the random variable X_i by lower case notation (i.e., by x_i).

To determine whether a decision maker's preferences satisfy the correct conditions so that we may use (1a) to capture her preferences, we need to define the concept of utility independence. When there are only two performance measures, attribute X_i is utility independent of X_j if preference for lotteries on X_i given $X_j = x_j$ does not depend on the level of x_j (Keeney and Raiffa 1976). The concept of utility independence allows us to consider the utility function for consequences of attribute X_i independent of X_j. Mutual utility independence holds when X_i is utility independent of X_j and X_j is utility independent of X_i. When there are more than two dimensions, mutual utility independence holds if for $X_1, X_2, \ldots X_n$, every subset of X_is is utility independent of its complement.

Given $\mathbf{X} = (X_1, X_2, \ldots X_n), n \ge 2$, if X_i is utility independent of X_j for all $j \ne i$, then the multilinear model (1a) is appropriate (Keeney and Raiffa 1976, p. 293). If mutual utility independence holds, then the correct choice is the multiplicative MAU model, $1 + wu(\mathbf{X}) = \prod_{i=1}^{n}[1 + ww_i u_i(X_i)]$. If we expand this compact form of the multiplicative model we get

$$u(\mathbf{X}) = \sum_{i=1}^{n} w_i u_i(X_i) + \sum_{i=1}^{n}\sum_{j>i} ww_i w_j u_i(X_i) u_j(X_j) + \sum_{i=1}^{n}\sum_{j>i}\sum_{m>j>i} w^2 w_i w_i w_m u_i(X_i) u_j(X_j) \times u_m(X_m) + \cdots + w^{n-1}\prod_{i=1}^{n} w_i u_i(X_1), \quad (1b)$$

where $0 \le w_i \le 1$ and $-1 < w < \infty$. Note there is no subscript on the common w, so this multiplicative form is a special case of the multilinear model (1a) where the strength of all interactions among criteria is the same. Finally, if a more restrictive preference condition called additive independence is satisfied (discussed later), we can represent the decision maker's preferences with an additive MAU model

$$u(\mathbf{X}) = \sum_{i=1}^{n} w_i u_i(X_i), \quad (1c)$$

where $0 \le w_i \le 1$ and $\sum_{i=1}^{n} w_i = 1$. Inspection of (1a), (1b), and (1c) reveals that the first term in both the multilinear and multiplicative models is the additive model. In other words, when the preferential interactions have no impact on preferences, i.e., when the interaction terms are all zero, the additive model is a special case of the more general aggregation schemes.

The additive model (1c) relies on a fairly strict assumption referred to as additive independence. Suppose a decision maker is faced with a choice between the following two alternatives:

A (x_{1*}, x_{2*}) with probability 0.5
(x_1^*, x_2^*) with probability 0.5

B (x_{1*}, x_2^*) with probability 0.5
(x_1^*, x_{2*}) with probability 0.5,

where x_{1*} and x_1^* represent the least and most preferred outcomes for measure i, respectively, and (y, z) is used to represent the joint receipt of y and z. Note that both A and B offer the best outcome on

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each measure with probability 0.5 and offer the worst outcome on each measure with probability 0.5. The difference between A and B is that with A you get either all best or all worst outcomes while with B you always get a mixture of best and worst. If the decision maker is indifferent between A and B, then essentially she is saying that it is the marginal, *not the joint*, distribution of each performance measure that determines preference. In other words, preference for one measure is independent of the level of the other measure, and we can simply add performance on each measure to arrive at an overall performance measure. This preference pattern is consistent with additive independence. When there are more than two measures, we can repeat the analysis for all pairs of measures.

It is easy to imagine situations where this restrictive assumption does not hold. As an example, Keeney and Raiffa (1976, p. 232) present the scenario faced by a farmer considering his preference for amounts of sunshine and rain. As Keeney and Raiffa indicate, in this situation it seems plausible that the amount of sunshine preferred by the farmer depends on the amount of rain that has fallen. However, additive models (1c) are generally quite robust (see, for example, Dawes and Corrigan 1974 and Dawes 1979), and typically provide a good approximation of preferences that do not satisfy additive independence. In other words, additive models usually identify the same top performing alternatives as aggregation models that do not satisfy additive independence. Further, Butler et al. (1997) provide a Monte Carlo approach that explores the impact of using a multiplicative (1b) aggregation model once an additive model (1c) has been assessed. It is straightforward to perform the same kind of analysis with a multilinear model (1a). Interested readers are directed to Keeney and Raiffa (1976) for a more detailed exposition on the applicable aggregation methods given less restrictive preference assumptions.

One popular form of single-attribute utility function is

$$u_i(x_i) = A_i - B_i e^{(x_i RT_i)} \tag{2}$$

(Clemen 1991, p. 379), where RT_i is the decision maker's assessed risk tolerance and A_i and B_i are scaling constants for measure i. The single-parameter exponential utility function in (2) is commonly used because it is flexible enough to model a wide variety of preferences and yet its parsimony and the fact that it is invertible add to the tractability of the analysis. The specific form of (2) is due to the decision maker's risk aversion ((2) is concave) and the fact that utility is decreasing in cost; the exponential utility function can also capture risk seeking (convex) utility. Several assessment techniques exist for eliciting utility functions from decision makers, i.e., for setting the parameters A_i, B_i, and RT_i (Logical Decisions 1996, p. 113). Figure 3 contains a graph of (2) for the utility of the cost of a project. See §6 for additional information on this utility function.

Several methods also exist for assigning weights to the performance measures (Schoemaker and Waid 1982, Logical Decisions 1996, p. 130). For example, a procedure called the trade-off method includes all n performance measures in $n-1$ pairwise trade-offs. In each trade-off the decision maker is asked to define hypothetical equally preferred alternatives (e.g., see Clemen 1991). Another popular weight assessment method is the analytical hierarchy process (Saaty 1988).

4. R&S Experimental Set-up

Assume that there are $K \geq 2$ project configurations. For $1 \leq k \leq K$, let $\boldsymbol{X}_k = (X_{k1}, X_{k2}, \ldots, X_{kn})$ denote a vector of random variables representing the performance measures for configuration k. Let $\mathrm{E}[u(X_k)]$ denote the expected utility (unknown) for configuration k and let

$$\mathrm{E}[u(\boldsymbol{X}_{[1]})] \leq \mathrm{E}[u(\boldsymbol{X}_{[2]})] \leq \cdots \leq \mathrm{E}[u(\boldsymbol{X}_{[K]})] \tag{3}$$

denote the ordered expected utility values. The goal is to select the project configuration with the largest expected utility $\mathrm{E}[u(\boldsymbol{X}_{[K]})]$. If the R&S procedure accurately identifies the configuration with the largest expected utility, we will say that a "correct selection" (CS) is made.

Because estimating the rank ordering in (3) reflects random fluctuation in configuration performance, it is impossible to guarantee a CS. Given this noise, if $\mathrm{E}[u(\boldsymbol{X}_{[K]})] \cong \mathrm{E}[u(\boldsymbol{X}_{[K-1]})]$, the final identification of $\mathrm{E}[u(\boldsymbol{X}_{[K]})]$ or $\mathrm{E}[u(\boldsymbol{X}_{[K-1]})]$ as the best configuration is less critical. Therefore, we ask the decision

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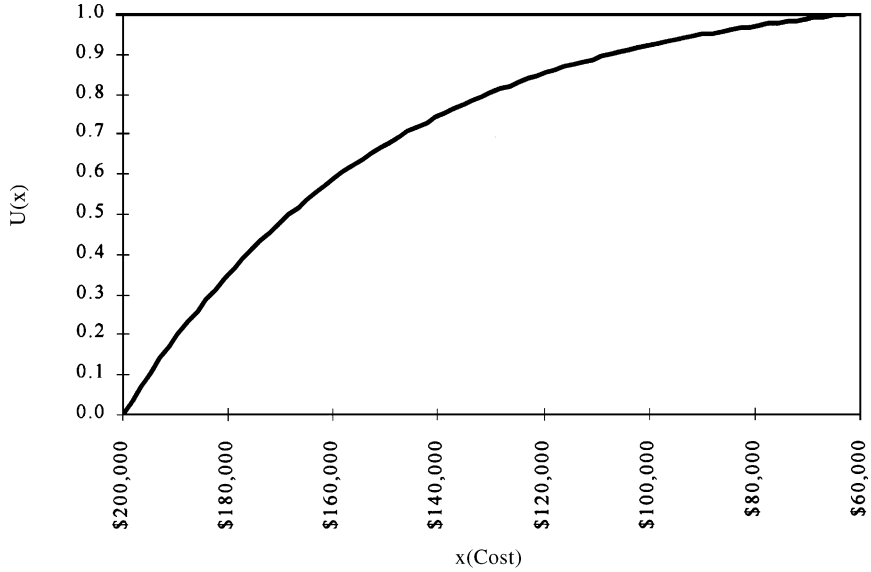
Figure 3 **Utility Function for Cost of Project with $A_i = 1.064$, $B_i = 0.0195$, and $RT_i = 50{,}000$**



maker to specify some level δ^* such that $\mathrm{E}[u(\boldsymbol{X}_{[K]})] - \mathrm{E}[u(\boldsymbol{X}_{[K-1]})] \geq \delta^*$ is practically significant. In the limit, the most difficult circumstances under which we will have to identify the best configuration is $\mathrm{E}[u(\boldsymbol{X}_{[1]})] = \mathrm{E}[u(\boldsymbol{X}_{[2]})] = \cdots = \mathrm{E}[u(\boldsymbol{X}_{[K-1]})] = \mathrm{E}[u(\boldsymbol{X}_{[K]})] - \delta^*$. When faced with this least favorable configuration and the inherent noise in the simulation process, the two-stage R&S procedure guarantees that the probability of a correct selection, $P\{\mathrm{CS}\}$, is greater than or equal to P^*, whenever $\mathrm{E}[u(\boldsymbol{X}_{[K]})] - \mathrm{E}[u(\boldsymbol{X}_{[K-1]})] = \delta^*$ for some P^* specified by the decision maker. In general, the R&S procedure is designed to satisfy the following probability requirement:

$$P\{\mathrm{CS}\} \geq P^* \text{ whenever } \mathrm{E}[u(\boldsymbol{X}_{[K]})] - \mathrm{E}[u(\boldsymbol{X}_{[K-1]})] \geq \delta^*, \tag{4}$$

where $(1/K) < P^* < 1$ and $0 < \delta^* < 1$. If $\mathrm{E}[u(\boldsymbol{X}_{[K]})] - \mathrm{E}[u(\boldsymbol{X}_{[K-1]})] < \delta^*$, then the procedure will select a configuration within δ^* of the best with probability at least P^*.

In our analysis, we will use the two-stage indifference-zone procedure for R&S due to Rinott (1978). In Stage 1, N_0 observations are sampled from each configuration, and we calculate the sample mean and variance, S_k^2, of $u(\boldsymbol{X}_k)$ for each configuration k. In Stage 2 we determine

$$N_k = \max\left\{N_0, \left\lceil \left(\frac{h}{\delta^*}\right)^2 S_k^2 \right\rceil\right\} \tag{5}$$

and sample $N_k - N_0$ additional observations from configuration k. The quantity h is a constant determined so that (4) is satisfied with equality for the least favorable configuration. Values for h given K, P^*, and N_0 can be found using tables or algorithms in Bechhofer et al. (1995, pp. 61–63). The notation $\lceil x \rceil$ represents the smallest integer $\geq x$. Upon completion of Stage 2, the configuration with the best sample mean performance over all N_k observations is selected. The approach presented here in no way relies on the admittedly conservative Rinott (1978) procedure, a feature we will try to highlight as the development unfolds.

In §6, we illustrate the Rinott (1978) procedure on our project example. Before we can use the Rinott procedure, we need a mechanism for specifying a level of practical significance, δ^*, which we address in the following section.

5. Selection of δ^*

In practice, the selection of δ^* depends on the judgment of the decision maker. Usually, it is assigned by

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appealing to the practical significance of the difference between the largest and second largest mean on a scale that has some physical meaning. For example, in the study conducted by Goldsman et al. (1991), four airline reservation systems were compared based on their expected time to failure. In this example, δ^* was selected on a time scale representing system time to failure. When R&S is based on expected utilities, the selection of δ^* can be challenging, because δ^* has no direct physical meaning on the utility scale. In this section, we discuss an approach to aid the decision maker in specifying δ^*.

5.1. A Utility Exchange Approach

In particular, we will select one criterion as the standard of measurement and exchange utility on the other performance measures for utility in the standard measure. For example, suppose that a decision maker is considering purchasing a car and is interested in two performance measures: cost and horsepower. The decision maker's choice problem for four hypothetical cars is presented in Table 1.

As demonstrated in Hammond et al. (1998), it is relatively straightforward to exchange utility in horsepower for utility in cost. Suppose we artificially set the horsepower of each car to a common level, say 140, and ask the decision maker to adjust each car's cost so that the "new" car is equally preferred to the original configuration. For example, the Harmony costs \$17,000 and comes equipped with 160 horsepower. If we were to decrease horsepower by 20, that would clearly be an inferior car and the decision maker would expect to pay less. Perhaps decreasing the horsepower from 160 to 140 would cause the decision maker to want to pay \$1,200 less, or \$15,800, for the Harmony. It should be noted that we are not stating that the 20 horsepower is worth \$1,200. The statement is dependent on the starting level of horsepower, in this case 160. Proceeding in a similar fashion, the decision maker is faced with a choice among the "new" cars in Table 2.

Now the decision maker's task has been simplified. Since all of the cars under consideration have the same level of horsepower, that criterion can be ignored when making the decision; it is no longer a discriminator among the alternatives. We can use the assessed weights and utility functions to formalize this intuitive procedure. At this point we would like to remind the reader that in the following methodological development, our arguments will be based on specific realizations of the random performance measures, which we represent as x_i.

It is important to note that when walking the decision maker through the swaps procedure that converts the original alternatives into equivalent hypothetical cars there is no need to make any assumptions about the form of the MAU aggregation function. In other words, the swaps procedure holds for all MAU, because the decision maker uses his or her internal utility function to provide the numbers required. We have outlined the swaps procedure to provide the intuition of the utility rescaling that we will outline in the following paragraphs, and the decision maker will never be required to make the swaps explicitly.

The first step in creating the hypothetical equally preferred alternatives is to select a measure as the medium for exchange or standard measure (e.g., cost in the previous example). Without loss of generality, let the standard measure be performance measure 1. Next, select a common level of utility c_i for the other criteria $i, 2 \le i \le n$ (e.g., 140 horsepower in the previous example). In other words, specify x'_{ki} such that $u_i(x'_{ki}) = c_i$ for all $i > 1$ and k, $1 \le k \le K$. The final step is to find the level of measure 1, x'_{k1}, such that the two alternatives are equally preferred.

Table 1 Alternatives by Measures Matrix for Car Selection

Alternative	Cost	Horsepower
Harmony	17,000	160
Starburst	17,000	125
Keyo	15,200	100
Palomino	18,500	160

Table 2 Equivalent Hypothetical Cars

Alternative	Cost	Horsepower
Harmony′	15,800	140
Starburst′	18,000	140
Keyo′	17,200	140
Palomino′	17,300	140

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PROPOSITION 1. *For the multilinear utility function* (1*a*), *an expression for computing the level of* x'_{k1} *is*

$$x'_{k1} = u_1^{-1}\left(\frac{u(\mathbf{x}_k) - Q_1}{Q_2}\right),$$

where

$$Q_1 = \sum_{i>1}^{n} w_i c_i + \sum_{i>1}^{n}\sum_{j>i} w_{ij} c_i c_j + \sum_{j>1}^{n}\sum_{j>i}\sum_{m>j>i} w_{ijm} c_i c_j c_m + \cdots \sum_{j>1}^{n}\sum_{j>1}\cdots \sum_{n-1>\cdots m>j>1} w_{ijm\cdots n-1} c_i c_j c_m \cdots c_{n-1} \quad (6)$$

and

$$Q_2 = w_1 + \sum_{j>1} w_{1j} c_j + \sum_{j>1}\sum_{m>j>1} w_{1jm} c_j c_m + \cdots + w_{123\cdots n} c_2 \cdots c_n. \quad (7)$$

PROOF. Set

$$u(\mathbf{x}'_k) = u(x'_{k1}, x'_{k2}, \ldots, x'_{kn}) = u(x_{k1}, x_{k2}, \ldots, x_{kn}) = u(\mathbf{x}_k).$$

The left-hand side can be rearranged to

$$u(\mathbf{x}'_k) = w_1 u_1(x'_{k1}) + \sum_{j>1} w_{1j} u_1(x'_{k1}) c_j + \sum_{j>1}\sum_{m>j>1} w_{1jm} u_1(x'_{k1}) c_j c_m + \cdots + w_{123\cdots n} u_1(x'_{k1}) c_2 \cdots c_n + Q_1.$$

We want to find $u_1(x'_{k1})$ such that

$$w_1 u_1(x'_{k1}) + \sum_{j>1} w_{1j} u_1(x'_{k1}) c_j + \sum_{j>1}\sum_{m>j>1} w_{1jm} u_1(x'_{k1}) c_j c_m + \cdots + w_{123\cdots n} u_1(x'_{k1}) c_2 \cdots c_n + Q_1 = u(\mathbf{x}_k)$$

or

$$u_1(x'_{k1})\left[w_1 + \sum_{j>1} w_{1j} c_j + \sum_{j>1}\sum_{m>j>1} w_{1jm} c_j c_m + \cdots + w_{123\ldots n} c_2 \cdots c_n\right] = u(\mathbf{x}_k) - Q_i,$$

$$u_1(x'_{k1}) = \frac{u(\mathbf{x}_k) - Q_1}{Q_2}. \quad (8)$$

The result follows from (8) by inverting the utility function. □

REMARK. This result also holds for the multiplicative (1b) and additive (1c) models. Because we will use it later in an example, (9) displays the relationship in (8) for the additive MAU model

$$u_l(x'_{k1}) = u_1(x_{k1}) + \sum_{i=2}^{n} \frac{w_i}{w_1}[u_i(x_{ki}) - c_i],$$

$$\text{or} \quad x'_{k1} = u_1^{-l}\left(u_1(x_{k1}) + \sum_{i=2}^{n} \frac{w_i}{w_1}\left[u_i(x_{ki}) - c_i\right]\right). \quad (9)$$

The quantity x'_{k1} is analogous to the rescaled costs presented in Table 2. In general $u_1(\cdot)$ may not be invertible. However, there is a wide array of invertible functional forms that are used in decision analysis (e.g., Clemen 1991). In this paper, we consider only exponential utility functions of the form in (2) that are, of course, invertible.

Care should be taken when selecting the common levels of performance, i.e., the $\{c_i\}$. It is possible that the exchange procedure in Proposition 1 will result in exchange utilities that are outside the range of the assessed utility function for Measure 1, i.e., $u_1(x'_{k1}) < 0$ or $u_1(x'_{k1}) > 1$ in (8). For the additive model in (1c), this problem can be avoided if a feasible solution in the $\{c_i\}$ exists to the following linear constraints, which come from (9):

$$u_1(x'_{kl}) = u_1(x_{k1}) + \sum_{i=2}^{n} \frac{w_i}{w_1}\left[u_i(x_{ki}) - c_i\right] \geq 0,$$

$$u_l(x'_{k1}) = u_1(x_{k1}) + \sum_{i=2}^{n} \frac{w_i}{w_1}\left[u_i(x_{ki}) - c_i\right] \leq 1, \quad (10)$$

$\forall k \in \{1, 2, \ldots, K\}$ and for all realizations of X_{k1} and X_{ki} for $2 \leq i \leq n$. If a feasible solution does not exist, then the decision maker can choose to extrapolate the utility function beyond the assessed range since having utility values between 0 and 1 is only a convention and is not necessary. If extrapolation is unacceptable, then the single-attribute utility function corresponding to the standard measure could be reassessed over a larger range and the exchange analysis performed again so that a feasible solution could be found to the constraints in (10). The $\{c_i\}$ selection procedure is similar but more complicated when either the multilinear (1a) or multiplicative (1b) models are used. The additional complexity follows from the fact that the set

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of constraints analogous to (10) is nonlinear for both models.

Since there are several choices for the common levels of the $\{c_i\}$, as previously discussed, one appealing choice is to set the level of all criteria $i, 2 \le i \le n$, to their most preferred level of performance, i.e., $c_i = 1, \forall i, 2 \le i \le n$. For the additive model in (1c), provided that this assignment is feasible in (10), the quantity $x_{k1} - x'_{k1}$ can be interpreted as the amount of Measure 1 on configuration k the decision maker is willing to exchange to shift performance on the other $i \neq 1$ criteria from their current to best levels of performance. This is particularly meaningful when cost is used as the standard of measurement, as we demonstrate in §6.2.

The utility exchange approach relies on the separability of preferences to convert multiple performance measures into a single measure of performance. Unlike an informal costing procedure, this approach relies on weights to reflect the relative importance of the criteria and utility functions to capture the marginal value of changing a performance measure. Since we have reduced the n criteria to a single measure of performance, the indifference zone is assessed on the criterion chosen as the measure of standardization.

After the utility exchange in (8), the R&S indifference-zone approach for the single-δ procedure is specified in the following manner. Let

$$\mathrm{E}\big[u_1\big(X'_{[1]1}\big)\big] \le \mathrm{E}\big[u_1\big(X'_{[2]1}\big)\big] \le \cdots \le \mathrm{E}\big[u_1\big(X'_{[K]1}\big)\big] \quad (11)$$

be the ordered expected utility values. The goal is to select the project configuration with the largest expected utility $\mathrm{E}[u_1(X'_{[K]1})]$. The R&S procedure is designed to satisfy the following probability requirement:

$$P\{CS\} \ge P^* \quad \text{whenever} \quad \mathrm{E}\big[u_1\big(X'_{[K]1}\big)\big] - \mathrm{E}\big[u_1\big(X'_{[K-1]1}\big)\big] \ge \delta_1^*, \quad (12)$$

where $(1/K) < P^* < 1$ and $0 < \delta_1^* < 1$.

In our procedure, we simulate to obtain realizations of $\boldsymbol{X}_k = (X_{k1}, X_{k2}, \ldots, X_{kn})$ for $k = 1, 2, \ldots, K$. Using (6), (7), and (8), we score $u_i(X'_{k1}) = (u(\boldsymbol{X}_k) - Q_1)/Q_2$ and then apply Rinott's procedure to the $u_1(X'_{k1})$ for $k = 1, 2, \ldots, K$. If the $u_1(X'_{k1})$ are not normally distributed as Rinott's procedure requires, then we obtain multiple ($M > 1$) realizations of $\boldsymbol{X}_k$, for each k, score each realization using (8), and compute an average for $u_1(X'_{k1})$ over the realizations for each k. In other words, each observation used in the Rinott procedure is the average of a batch of $M > 1$ individual replication results. Goldsman et al. (1991) refer to this averaging step as making macroreplications.

5.2. Validity of the Utility Exchange Approach

The following two propositions establish the equivalence of the R&S procedure stated in Expressions (3) and (4) with the R&S procedure stated in Expressions (11) and (12). Note that the results apply for the multilinear, multiplicative, and additive models. Because the other models are special cases of the multilinear, we use this form in the proofs and provide the specific results for the additive model, because we will use it in an example in §6.

Proposition 2. *The following relationship holds*:

$$\mathrm{var}\big(u_1\big(X'_{k1}\big)\big) = \frac{\mathrm{var}(u(\boldsymbol{X}_k))}{Q_2^2} \quad (13)$$

for the standard measure 1 *and configuration* $k, k = 1, 2, \ldots, K$, *for MAU function* (1*a*).

Proof. From (8)

$$\mathrm{var}\big(u_1\big(X'_{k1}\big)\big) = \mathrm{var}\left(\frac{u(\boldsymbol{X}_k) - Q_1}{Q_2}\right),$$

which yields (13) since Q_1 and Q_2 are constants. □

Remark. For the additive MAU (1c), the relationship in (13) reduces to

$$\mathrm{var}(u_1(X'_{k1})) = \frac{\mathrm{var}(u(\boldsymbol{X}_k))}{w_1^2}.$$

Proposition 3. $\mathrm{E}[u(X_{[K]})] - E[u(X_{[K-1]})] \ge \delta^*$ *iff* $\mathrm{E}[u_1(X'_{[K]1})] - \mathrm{E}[u_1(X'_{[k-1]1})] \ge \delta/Q_2 = \delta_1^*$.

Proof. From (8) we know that $\mathrm{E}[u(X_k)] = Q_2\mathrm{E}[u_1(X_{k1})] + Q_1$ for the multilinear model. Therefore,

$$\begin{aligned}\delta^* &\le \mathrm{E}[u(\boldsymbol{X}_{[K]})] - \mathrm{E}[u(\boldsymbol{X}_{K-1})] \\ &= Q_2\mathrm{E}\big[u_1\big(X'_{[K]1}\big)\big] + Q_l - \big(Q_2\mathrm{E}\big[u_1\big(X'_{[K-1]1}\big)\big] + Q_1\big) \\ &= Q_2\big(\mathrm{E}\big[u_1\big(X'_{[K]1}\big)\big] - \mathrm{E}\big[u_1\big(X'_{[K-1]1}\big)\big]\big)\end{aligned}$$

and the result follows. □

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Remark. For the additive MAU (1c), Proposition 2 is $\mathrm{E}[u_1(X'_{[K]1})] - \mathrm{E}[u_1(X'_{[K-1]1})] \geq \delta^*/w_1 = \delta_1^*$.

In the Rinott (1978) procedure, the sample size for the second stage is a function of the ratio of the standard deviation and the indifference-zone parameter. In light of Propositions 2 and 3, this has two implications. First, if the indifference-zone parameter δ^* in Expression (4) is chosen to be equal to $Q_2\delta_1^*$, then the two R&S procedures are completely equivalent. Hence the problem of assessing the R&S procedure and assessing δ^* on the MAU function reduces to the problem of setting up the R&S procedure and assessing δ_1^* on the single-attribute utility function corresponding to the standard performance measure. The second implication is that the R&S procedure based on the standard performance measure is invariant under the choice of the $\{c_i\}$ since Q_2 cancels out of the ratio of the standard deviation and the indifference-zone parameter.

5.3. Establishing the Indifference Zone

To establish an indifference zone, we recommend assessing δ_1^* in the units of the performance measure and then converting to a number on the utility, (0,1), scale. For example, one can provide the decision maker with the following scenario: "If Projects A and B are identical on all criteria except cost and Project A costs \$500,000, what is the maximum cost of B that could cause you to feel that B is significantly better than A?" Then $\delta_1^* = u(\text{Cost of } B) - u(\text{Cost of } A)$. The assessment process should be iterative, varying the cost of project A to ensure that the decision maker's specification of δ_1^* does not depend on the reference cost of Project A. Additionally, an indifference-zone–preference-zone diagram on the performance measure (Bechhofer et al. 1995, p. 178) could be used to aid the assessment and to check for the consistency of the decision maker's preferences.

To construct an indifference-zone–preference-zone diagram on the performance measure, we use *certainty equivalents* defined on the single-attribute utility functions. For a single-attribute utility function, the certainty equivalent is equal to the inverse of the utility function evaluated at the expected utility (Clemen 1991, p. 372). Let $CE_{[K]1}$ denote the certainty equivalent corresponding to $\mathrm{E}[u_1(X_{[K]1})]$. Then, by definition,

$$\mathrm{E}[u_1(X_{[K]1})] = u_1(CE_{[K]1}). \tag{14}$$

From (12) and (14), the indifference zone is defined by $u_1(CE_{[K]1}) - u_1(CE_{[K-1]1}) \geq \delta_1^*$. To establish the indifference-zone–preference-zone on the original performance measure, one can invert $u_1(CE_{[K]1})$ and $u_1(CE_{[K-1]1})$ and establish an indifference zone based on $CE_{[K]1}$ and $CE_{[K-1]1}$. As previously mentioned, the single-attribute utility function may not be invertible. However, in this paper we assume the exponential form that guarantees inversion.

It is important to note that for a constant indifference-zone parameter value δ_1^* on the utility measure, the indifference zone on the original performance measure will be variable unless the utility function is linear. We are not aware of any other work in R&S that defines a variable indifference zone of this sort. Therefore, this may be viewed as an important embellishment of R&S. To demonstrate, consider the indifference-zone–preference-zone for $CE_{[K]1}$ and $CE_{[K-1]1}$ when the utility function in (2) is used. The curve dividing the indifference zone from the preference zone is constructed by setting $u_1(CE_{[K]1}) - u_1(CE_{[K-1]1}) = \delta_1^*$ and solving for $CE_{[K]1}$. The resultant expression for the curve dividing the indifference zone from the preference zone is

$$CE_{[K]1} = CE_{[K-1]1} + RT_1 \ln\left\{-\left(\frac{\delta_1^*}{B_1}\right) e^{\frac{-CE_{K-1]1}}{RT_1}}\right\}.$$

Figure 4 contains three indifference-zone–preference-zone diagrams corresponding to δ_1^* equal to 0.01, 0.1, and 0.2, respectively, for the cost-utility function example in Figure 3. The preference zone always appears below the curve and the indifference zone above as indicated on the graphs because a lower cost is better. As δ_1^* decreases the indifference-zone parameter on the performance measure axis decreases and tends toward a constant value across its entire range. This explains why the relationship becomes more linear as δ_1^* decreases.

5.4. Characterizing the Indifference Zone for All Performance Measures

Once δ_1^* has been determined, Proposition 3 can also be used for computing the implied δ_i^* for $2 \leq i \leq n$,

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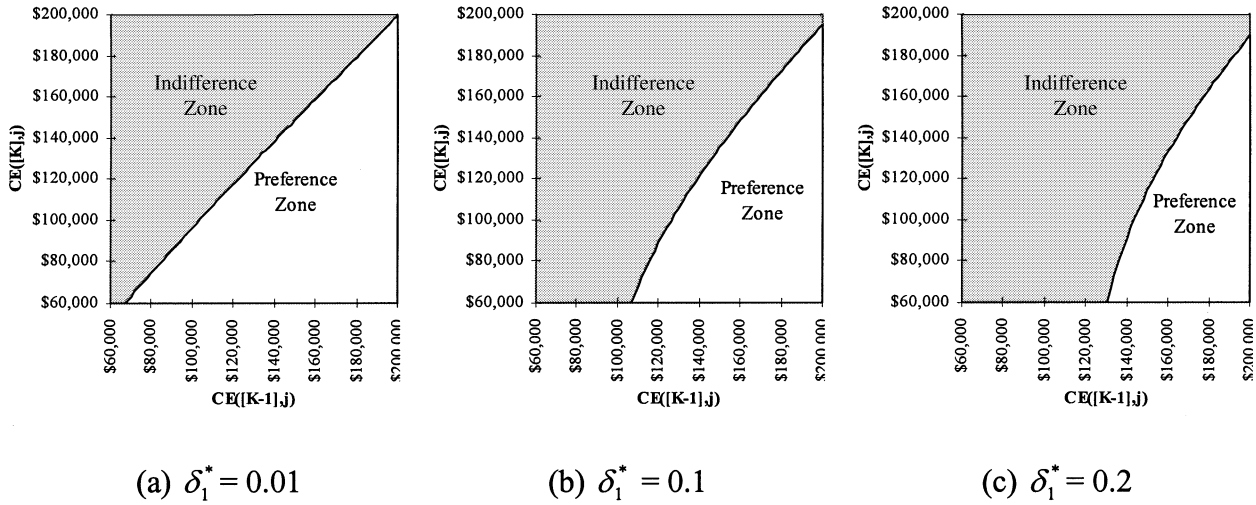
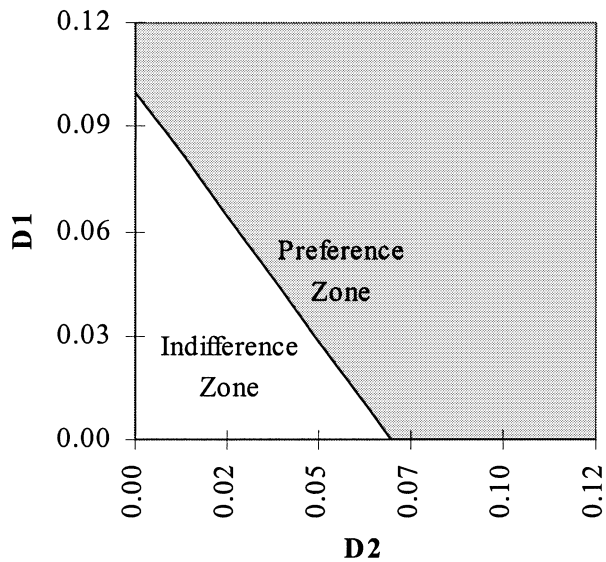
Figure 4 Indifference-Zone–Preference-Zone Curves for Cost



if the ith performance measure were used as the standard. This follows since the weights are assumed to be known from the MAU assessment procedure, δ_1^* can be used to compute δ^*, and then can be used to compute the other δ_i^*. By inverting the $u_i(\cdot)$, one can construct indifference-zone–preference-zone diagrams for all the other original performance measures implied by δ_1^*, the weights, and the single-attribute utility functions. Again indifference-zone–preference-zone diagrams are helpful to check for consistency of the decision makers' preferences expressed in the weights and single-attribute utility curves.

Additionally, Propositions 1 and 3 can be used to derive the multiple-dimension indifference zone across all performance measures. To illustrate, consider the case of the additive model in (1c). Since (9) is linear, the contribution to the mean utility difference between any two configurations corresponding to any performance measure is linear. For the standard measure, the contribution coefficient is one; for performance measure i, the contribution coefficient is w_i/w_1 for $2 \le i \le n$. For example, suppose that $n = 2$, Measure 1 is selected as the standard measure, δ_1^* is assessed to be 0.1, and $w_1 = 1 - w_2 = 0.4$. Figure 5 depicts the indifference zone for this two-dimensional example. The vertical axis label $D1$ represents $\mathrm{E}[u_1(X_{[K]1})] - \mathrm{E}[u_1(X_{[K-1]1})]$ and the horizontal axis label $D2$ represents $\mathrm{E}[u_2(X_{[K]2})] - \mathrm{E}[u_2(X_{[K-1]2})]$. The line separating the indifference zone from the preference zone is $D1 = 0.1 - (3/2)D2$. In higher dimensions, the indifference zone is separated from the preference zone by a hyperplane whose form is defined by (9) and the assessed δ_1^*. For the models in (1a) and (1b), the functions dividing the indifference zone from the preference zone (e.g., (8) for the model in (1a)) would not be linear, since (1a) and (1b) contain interaction terms in the single-attribute utility functions.

Figure 5 Example of an Indifference-Zone–Preference-Zone in Two Dimensions



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6. Application of the Procedures

In this section, we illustrate our methodology on an example. Although the data used in the example are not real, they are representative. The simulator generated results on a land seismic survey that is realistic in both size and structure. Additionally, the utility functions and weights were assessed based on informal discussions with personnel who have field management experience. From these discussions, we determined that the additive model in (1c) was sufficient. Therefore, we use (1c) throughout this section.

6.1. Experimental Set-up

We define the configurations based on resource levels along two dimensions: the number of source crews and the number of units of receiving equipment. Resource decisions along these two dimensions are considered the most important on a land seismic survey. We consider four configurations: one and two source crews in conjunction with 1,100 and 1,300 units of receiving equipment. All other resources and parameters remain fixed.

The performance measures include survey cost, survey duration, and utilization for the following four crews: source, layout, transport, and packing. One might argue that survey cost and duration are redundant. Indeed, they are highly positively correlated performance measures. However, a decision maker's preferences for cost and duration may differ because cost may not capture such intangibles as loss of reputation from not finishing a job on time. Such preferences could be captured in the utility assessment for job duration. The recording crew utilization is not included in the set of performance measures, because it rarely bottlenecks production. Additionally, signal crews and recording crews work in such a synchronized fashion that we treat them as a single entity in this analysis.

The utility function for the survey cost and survey duration were defined over a range considered reasonable for a survey of the given size and complexity. Specifically, survey cost was defined over the range of \$60,000 to \$200,000 with the following utility function previously shown in Figure 3:

$$1.064 - (0.0195)e^{(x/50000)}.$$

Job duration was defined over the range 240 to 550 hours with the following utility function:

$$1.021 - (0.00106)e^{(x/80)}.$$

The utility functions for the crew utilizations were all defined over the range 0 to 1.0. Utility functions for utilization must capture worker satisfaction. We do this by creating a utility function that remains high until the utilization reaches a certain level. Beyond this point utility drops off rapidly to model worker dissatisfaction from being overworked. These aspects are captured in the following utility function:

$$1 - (2.061E - 09)e^{(x/0.05)}.$$

The same utility function was used for utilization on all four crews. Note that for each of the above utility functions, the risk tolerance parameters were selected based on discussions with experienced field personnel.

The MAU function was constructed from a weighted sum of the six single-attribute utility functions. Weights were assessed as follows: cost (0.4), job duration (0.2), and worker satisfaction for each utilization (0.1).

6.2. Results

We chose cost as the measure of standardization (i.e., Measure 1). For the Rinott (1978) two-stage procedure, we used the following parameters: $\delta_1^* = 0.00434$, $P^* = 0.75$. Recall that for the assessment of δ_1^*, any two points on the performance measure scale can be used. We followed the procedure described in §5.3 and asked the decision maker to consider the following scenario: Configurations A and B are identical on all measures except cost. If Configuration A is expected to cost \$120,000, what is the maximum cost of Configuration B that would cause you to conclude that B is significantly better than A? The decision maker responded that the maximum cost alluded to would be \$119,000. Using the utility function for cost displayed in Figure 3, $\delta_1^* = u(119{,}000) - u(120{,}000) = 0.00434$. Finally, we decided to use the best level of performance as the common level for the other measures, i.e., $c_i = 1$ for all $i \neq 1$ (i.e., the cost measure).

In the first stage of the Rinott procedure, 100 simulation runs were made for each configuration.

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Table 3 First-Stage Results from the R&S Procedure

Configuration	Average Rescaled Utility	Rescaled Standard Deviation	Additional Macro-replications
I	0.571	0.0102	15
II	0.960	0.0011	0
III	0.542	0.0108	18
IV	0.861	0.0036	0

Table 4 Second-Stage Results from the R&S Procedure

Configuration	Average Rescaled Utility	Rescaled Standard Deviation	Average Cost	Average Equivalent Cost
I	0.5759	0.0113	$112,354	$160,947
II	0.9595	0.0011	$ 68,551	$ 84,261
III	0.5403	0.0102	$151,197	$164,490
IV	0.8607	0.0036	$ 93,507	$117,275

Since the rescaled utilities were not normally distributed, averages of the values were calculated based on batches of size 10 (i.e., $M = 10$), yielding 10 macroreplications for each configuration. The hypothesis of normality was not rejected for any of the samples based on the chi-square, Kolmogorov-Smirnov, and Anderson-Darling tests in *BestFit* (Palisade 1996).

Table 3 contains the results from the first stage of the R&S procedure. The configurations are numbered as follows: a single-source crew with 1,100 units of equipment (I), a single-source crew with 1,300 units of equipment (II), two source crews with 1,100 units of equipment (III), and two source crews with 1,300 units of equipment (IV). The number of required additional macroreplications was calculated using the formulae and tables in Bechhofer et al. (1995, pp. 61–63).

At first glance Table 3 appears counterintuitive. It appears obvious that Configuration II is superior based on its relatively high average rescaled utility and the small standard deviation of all configurations. It is important to note that the Rinott (1978) procedure is quite conservative. In fact, examination of (5) reveals that the Stage-2 mechanism for determining the number of additional simulations is based solely on the variance (standard deviation) of the sample estimates; in no way do the mean levels of performance or the differences in mean levels enter into this specification. Note that the larger the standard deviation in Table 3, the more additional required simulations. This conservatism is the result of the Rinott (1978) procedure and not the proposed MAU R&S procedure. We will address this issue more fully in the conclusion.

Table 4 contains second-stage results, and Configuration II is best since it has the highest sample average. These results reveal that 200 additional units of equipment are more beneficial than an additional source crew. Adding both together is not worth the additional cost. A closer inspection of the data reveals that with 1,100 units of equipment, equipment is the bottleneck. Therefore, adding an additional source crew without extra equipment does not provide any improvement. In fact, the cost of an additional source crew becomes a detriment (compare Configurations I and III), and the average rescaled utility decreases. With 1,300 units, the layout crew becomes the bottleneck (average utilization of about 94%). Once again, adding the additional source crew increases cost without increasing productivity and average rescaled utility decreases (compare Configurations II and IV).

Additionally, we examine the last two columns of Table 4; from left to right, the average "real" simulated cost from the full set of replications, $\overline{X}_{\text{cost}}$, and mean equivalent cost, i.e., (9), respectively. We can now discuss the difference in the configurations' performance in terms of dollars. For example, Configuration II is the most preferred by, on average, approximately 33,000 (117,275–84,261) equivalent dollars. Further, we can provide some intuition as to how good a particular configuration is and provide more insight as to why some configurations outperform others.

As an illustration, compare Configurations II and III. Configuration II is clearly the superior allocation based on the rescaled utility and equivalent cost, which should, and do, provide consistent results. What is interesting is to look at the differences between equivalent cost and average cost. Recall that we performed the rescaling by arbitrarily setting the common level of the noncost criteria to their best level. Thus, equivalent cost–average cost is the amount the decision-maker should be willing to pay to shift the performance on the noncost criteria from

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their actual to the best level of performance. Intuitively, the better a configuration performs, the less one should be willing to pay to move performance to the most preferred level; in the limit, one would pay \$0 to improve a configuration that was already performing at the best level.

Examining Configuration III, the decision maker would be willing to exchange about \$13,300 (164,490–151,197) to shift performance on noncost criteria to their most preferred level. For Configuration II, the same argument yields an exchange price of about \$15,700 (84,261–68,551). This implies that Configuration III is a little closer to the best level of performance on the *noncost* criteria than Configuration II. However, this gain in performance is very expensive; the average cost of Configuration III is about 35% as higher than the average cost for Configuration II. Given the relatively high weight on cost, the gains in noncost performance of Configuration III are offset by its high cost, and Configuration II is the most preferred.

6.3. Sensitivity Analysis of the Weights

Based on Table 4, it appears that Configuration II is the most preferred alternative. The next step in most applications of multiattribute utility theory would focus on a series of sensitivity analyses to establish a sense of the robustness of the recommendations generated by the MAU model (e.g., Dyer et al. 1998). One common technique is to determine the impact of varying the assessed weights of the MAU model. Often, this analysis is performed by varying one weight at a time while the ratios among the other weights are held constant. Morrice et al. (1999) use this approach to study weight sensitivity for the land seismic survey application discussed in this paper. However, varying one weight at a time can be misleading because it ignores the potential interaction that can result from simultaneous manipulations of multiple weights (e.g., Butler et al. 1997).

This concern can be alleviated by evaluating all possible combinations of weights via Monte Carlo simulation (Butler et al. 1997). The first step in randomly generating weights for n attributes is to select $n-1$ independent random numbers from a uniform distribution on (0, 1). The second step is to rank these numbers. Suppose the ranked numbers are $1 > r_{n-1} \geq \cdots \geq r_{(2)} \geq r_{(1)} > 0$. The first differences of these ranked numbers (including the bounds 0 and 1) can be obtained as $w_n = 1 - r_{(n-1)}$, $w_{n-1} = r_{(n-1)} - r_{(n-2)}, \ldots,$ $w_1 = r_{(1)} - 0$. Then, the set of numbers $(w_1, w_2, \ldots, w_n)$ will sum to one and $(w_1, w_2, \ldots, w_{n-1})$ is uniformly distributed over the simplex $(\{y_1, y_2, \ldots, y_{n-1}\}: y_i \geq 0, \sum_{i=1}^{n-1} y_i \leq 1)$ (Devroye 1984, Theorem 2.1, p. 207). This result can be used to rapidly generate weights via Monte Carlo simulation.

As demonstrated in Butler et al. (1997), these "random weights" can be useful in sensitivity analysis. However, in this case we chose to meet with decision makers to determine what relationships among the weights were consistent with their preferences. After some discussion we concluded that even if the assessed weights were not precise, we were confident that cost was the most important criteria followed by job duration; in other words, the weight on cost was at least as large as the weight on job duration. Further, while job duration was relatively more important than all four of the crew utilization measures, it was difficult to determine which of the utilization measures was most important. We were able to use this information in a Monte Carlo simulation sensitivity analysis of the weights.

Table 5 displays the results of 1,000 simulated weight sets that satisfy the decision maker's importance ranking of the criteria. For each of the 1,000 iterations, we generated a set of six random weights as previously described. The largest of each set of weights was assigned to the project cost, the second largest assigned to duration, and one fourth of the remaining weight was assigned to each of the crew utilization measures. This weight scheme was selected to reflect the notion that the weights on cost and duration were quite a bit more important than the utilization measures, and it is difficult to specify which utilization measure is relatively more important. The rows in Table 5 summarize the ranking distribution of each alternative.

As Table 5 indicates, the recommendation of Configuration II is *extremely* robust under these weight assumptions. In fact, it is the top-ranked alternative for all 1,000 simulated weight sets. The rightmost four columns of Table 5 display the total number (for both

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Table 5 Sensitivity Analysis to Weights

	Configuration Rank				Total Number of Simulations Required for Configuration			
	I	II	III	IV	I	II	III	IV
Min	3	1	3	2	100.0	100.0	210.0	100.0
5th percentile	3	1	3	2	100.0	100.0	230.0	100.0
25th percentile	3	1	3	2	220.0	100.0	270.0	100.0
50th percentile	4	1	3	2	390.0	100.0	310.0	100.0
75th percentile	4	1	4	2	590.0	100.0	350.0	100.0
95th percentile	4	1	4	2	780.0	100.0	400.0	120.0
Max	4	1	4	2	820.0	100.0	400.0	290.0
Average	3.60	1.00	3.40	2.00	412.2	100.0	312.3	103.0
SD	0.49	0.00	0.49	0.00	216.4	0.0	51.4	13.9

stages) of simulations required by the Rinott procedure. Finally, we refer the reader to Butler et al. (1997) for a related sensitivity analysis that allows the analyst to test the model recommendations sensitivity to the assumption of an additive MAU model.

7. Conclusion

In this paper we have developed an R&S procedure applied to multiple project configurations that are evaluated on multiple performance measures. The core procedure relies on the ideas and techniques found in MAU theory. Our example demonstrates that it can be applied to realistic problems in which simulation is used. The use of multiattribute utility theory essentially provides a formal mechanism with which a decision maker can "cost out" performance in one criteria for performance in another. Thus the approach provides a formal procedure for an intuitive strategy. Care should be taken when evaluating these rescaled "costs." While the costs are relevant for decision making, they do not represent any real costs that the decision maker should expect to incur.

In this presentation we focused on the Rinott (1978) procedure for ease of exposition. This generalized R&S procedure presented here is in no way dependent on the use of this conservative procedure. The utility exchange procedure presented in §5.1 results in a scalar performance measure that summarizes performance on multiple criteria, including nonlinear preference functions and the relative importance of the measures. After the swaps are performed and we have a scalar, any existing ranking and selection approach can be utilized to select the top performing alternative(s). For example, in future research it would be natural to combine this work with the work of Chen et al. (1998, 2000a,b) that controls the computing budget allocation over the different configurations. The weights add another dimension to this analysis since they affect the relative importance of a performance measure, its impact on the variation, and thus its impact on the computing budget. A related idea is presented by Chick and Inoue (1998, 2001), who use a decision theoretic approach that uses both the desired probability of a correct selection and the cost of replication to determine the number of additional replications required to reduce the risk of an incorrect selection. While their approach focuses on a single criterion, one could extend our methodology to include their Bayesian technique and potentially relieve some of the computational burden associated with all R&S procedures. Finally, it would be interesting to combine our work with R&S procedures designed to facilitate variance reduction through the use of common random numbers (see Matejcik and Nelson 1995 and Goldsman and Nelson 1998).

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